Tidy Tuesday 3/31

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2026-03-31

```r
# Using R
# Option 1: tidytuesdayR R package
## install.packages("tidytuesdayR")

tuesdata <- tidytuesdayR::tt_load('2026-03-31')
```

```
## ---- Compiling #TidyTuesday Information for 2026-03-31 ----
## --- There are 2 files available ---
##
##
## -- Downloading files --------------------------------------------------------
##
##   1 of 2: "ocean_temperature.csv"
##   2 of 2: "ocean_temperature_deployments.csv"
```

```r
## OR
tuesdata <- tidytuesdayR::tt_load(2026, week = 13)
```

```
## ---- Compiling #TidyTuesday Information for 2026-03-31 ----
## --- There are 2 files available ---
##
##
## -- Downloading files --------------------------------------------------------
##
##   1 of 2: "ocean_temperature.csv"
##   2 of 2: "ocean_temperature_deployments.csv"
```

```r
ocean_temperature <- tuesdata$ocean_temperature
ocean_temperature_deployments <- tuesdata$ocean_temperature_deployments

# Option 2: Read directly from GitHub

ocean_temperature <- readr::read_csv('https://raw.githubusercontent.com/rfordatascience/tidytuesday/mai
```

```
## Rows: 19165 Columns: 5
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## dbl  (4): sensor_depth_at_low_tide_m, mean_temperature_degree_c, sd_temperat...
## date (1): date
```

```
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
ocean_temperature_deployments <- readr::read_csv('https://raw.githubusercontent.com/rfordatascience/tid
```

```
## Rows: 14 Columns: 5
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (1): deployment_id
## dbl  (2): latitude, longitude
## date (2): start_date, end_date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

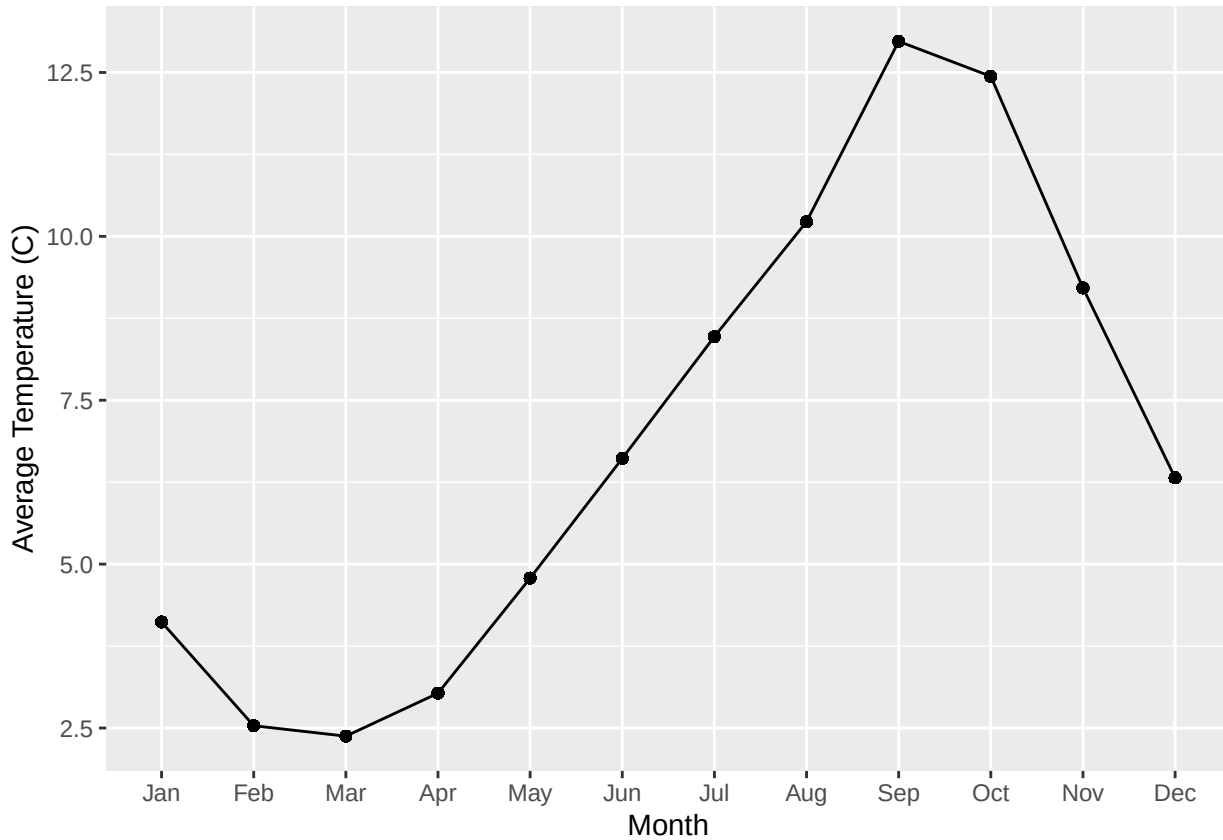
```
df1 <- ocean_temperature %>%
  mutate(month = month(date, label = TRUE), year = year(date), day = day(date)) %>%
  group_by(month) %>%
  mutate(avg_temp = mean(mean_temperature_degree_c))

df1
```

```
## # A tibble: 19,165 x 9
## # Groups:   month [12]
##    date       sensor_depth_at_low_tide_m mean_temperature_degree_c
##    <date>                          <dbl>                     <dbl>
##  1 2018-02-20                          2                      1.58
##  2 2018-02-20                          5                      1.50
##  3 2018-02-20                         10                      1.49
##  4 2018-02-20                         15                      1.63
##  5 2018-02-20                         20                      1.84
##  6 2018-02-20                         30                      1.92
##  7 2018-02-20                         40                      1.82
##  8 2018-02-21                          2                      1.72
##  9 2018-02-21                          5                      1.46
## 10 2018-02-21                         10                      1.42
## # i 19,155 more rows
## # i 6 more variables: sd_temperature_degree_c <dbl>, n_obs <dbl>, month <ord>,
## #   year <dbl>, day <int>, avg_temp <dbl>
```

```
plot1 <- ggplot(df1, aes(x = month, y = avg_temp, group = 1)) +
  geom_line() +
  geom_point() +
  labs(x = "Month", y = "Average Temperature (C)")

plot1
```

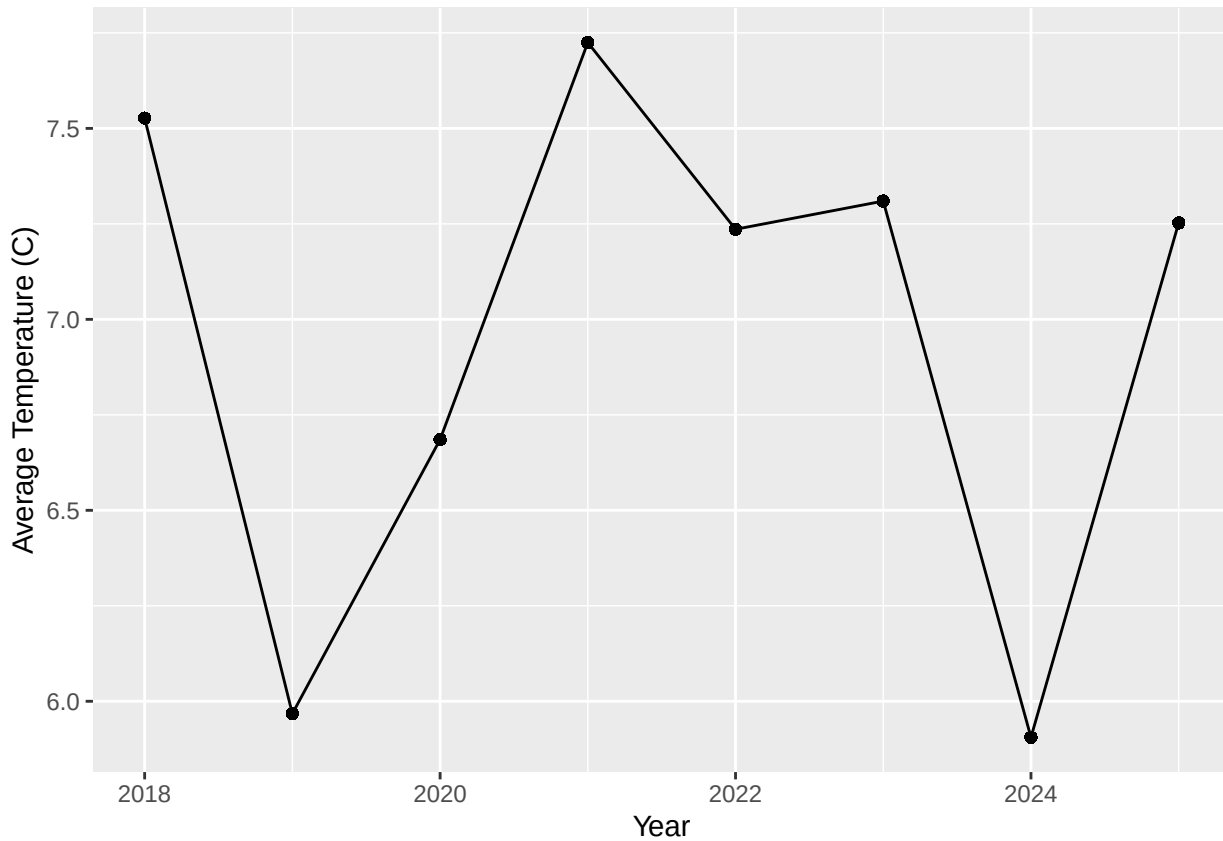



```
df2 <- ocean_temperature %>%
  mutate(month = month(date, label = TRUE), year = year(date), day = day(date), avg_temp = mean(mean_te
  group_by(year) %>%
  mutate(avg_temp = mean(mean_temperature_degree_c))
df2
```

```
## # A tibble: 19,165 x 9
## # Groups:   year [8]
##    date       sensor_depth_at_low_tide_m mean_temperature_degree_c
##    <date>                          <dbl>                     <dbl>
##  1 2018-02-20                          2                      1.58
##  2 2018-02-20                          5                      1.50
##  3 2018-02-20                         10                      1.49
##  4 2018-02-20                         15                      1.63
##  5 2018-02-20                         20                      1.84
##  6 2018-02-20                         30                      1.92
##  7 2018-02-20                         40                      1.82
##  8 2018-02-21                          2                      1.72
##  9 2018-02-21                          5                      1.46
## 10 2018-02-21                         10                      1.42
## # i 19,155 more rows
## # i 6 more variables: sd_temperature_degree_c <dbl>, n_obs <dbl>, month <ord>,
## #   year <dbl>, day <int>, avg_temp <dbl>
```

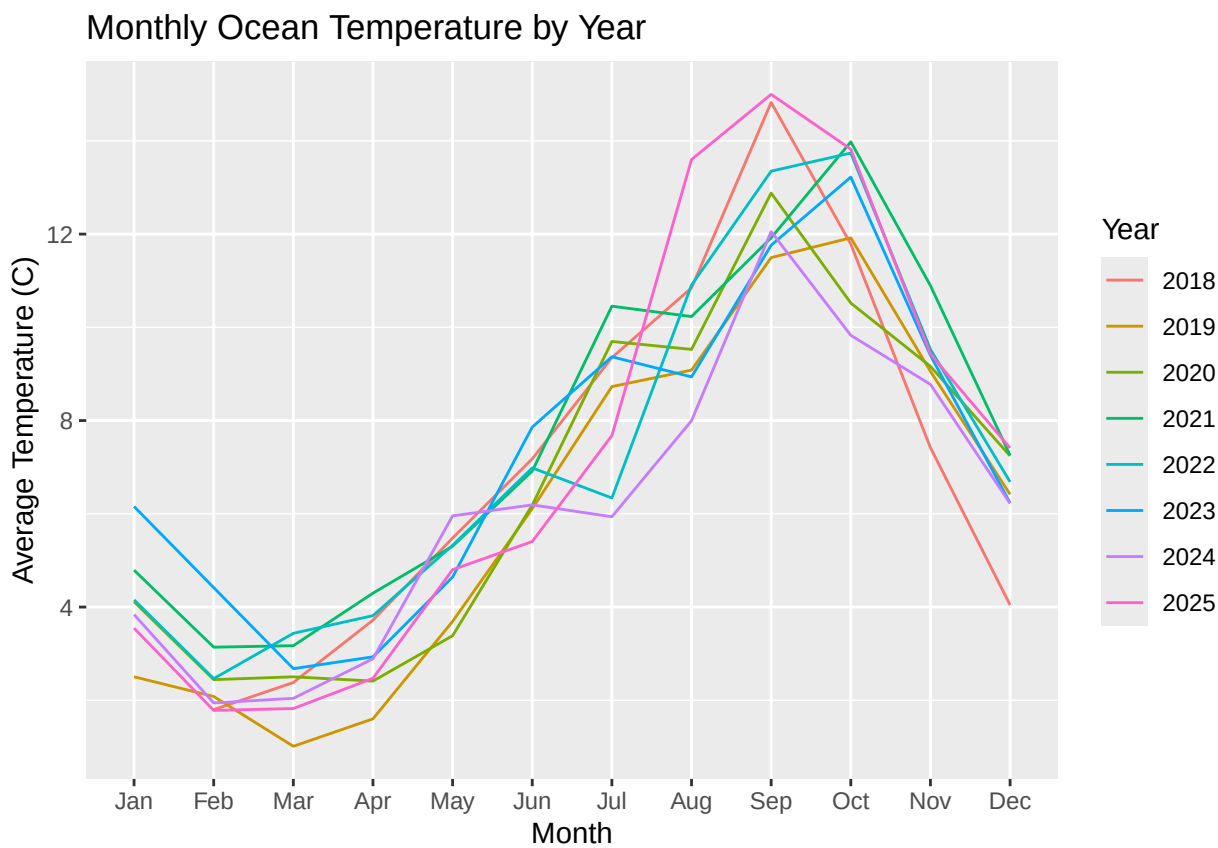
```r
plot2 <- ggplot(df2, aes(x = year, y = avg_temp)) +
  geom_line() +
  geom_point() +
  labs(x = "Year", y = "Average Temperature (C)")

plot2
```




```r
df3 <- ocean_temperature %>%
  mutate(
    month = month(date, label = TRUE),
    year = year(date)
  ) %>%
  group_by(year, month) %>%
  mutate(avg_temp = mean(mean_temperature_degree_c))

ggplot(df3, aes(x = month, y = avg_temp, group = year, color = factor(year))) +
  geom_line() +
  labs(
    title = "Monthly Ocean Temperature by Year",
    x = "Month",
    y = "Average Temperature (C)",
    color = "Year"
  )
```

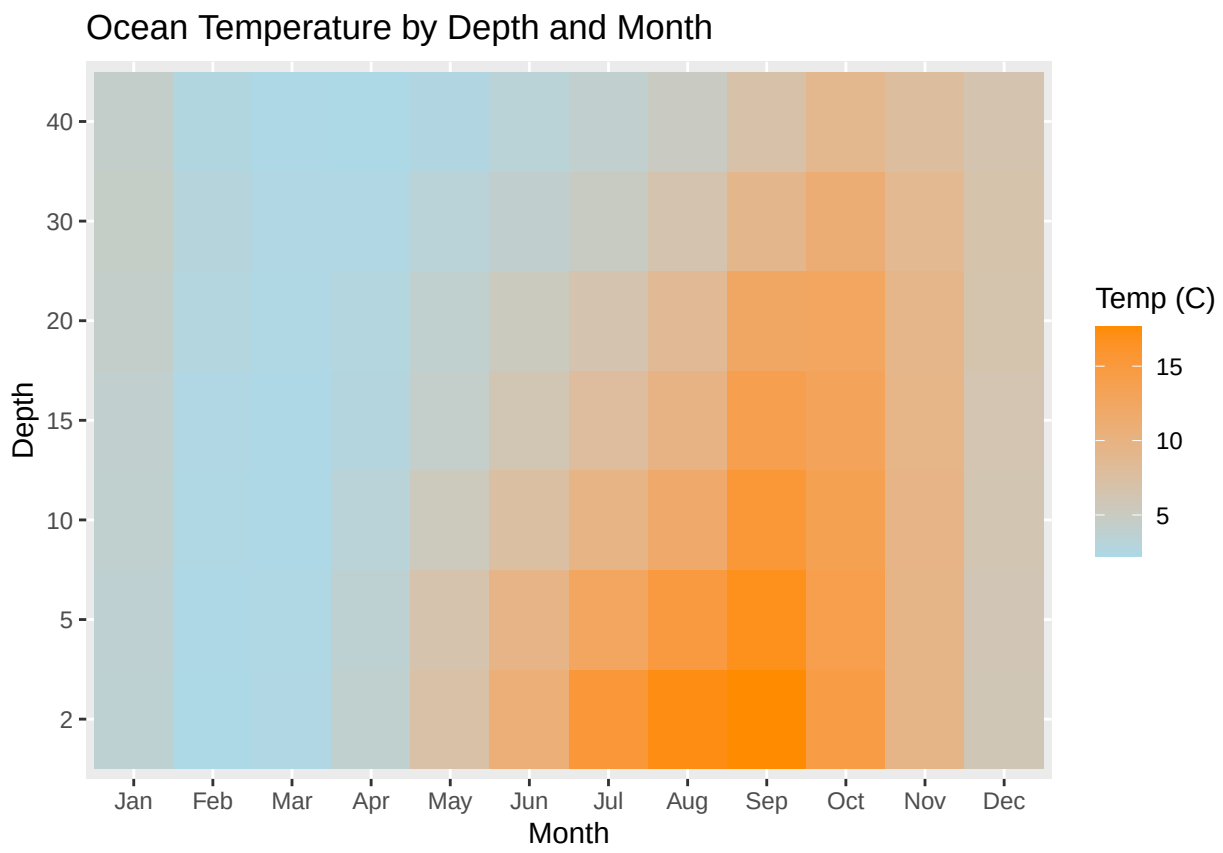



```r
#How does temperature change with depth? Does the relationship change over time?

df4 <- ocean_temperature %>%
  mutate(
    month = month(date, label = TRUE, abbr = TRUE)
  ) %>%
  group_by(month, sensor_depth_at_low_tide_m) %>%
  summarise(avg_temp = mean(mean_temperature_degree_c, na.rm = TRUE), .groups = "drop")

df4 <- df4 %>%
  mutate(depth = factor(sensor_depth_at_low_tide_m, levels = c(2, 5, 10, 15, 20, 30, 40)))

ggplot(df4, aes(x = month, y = depth, fill = avg_temp)) +
  geom_tile() +
  labs(
    title = "Ocean Temperature by Depth and Month",
    x = "Month",
    y = "Depth",
    fill = "Temp (C)"
  ) +
  scale_fill_gradient(low = "lightblue", high = "darkorange")
```


Ocean Temperature by Depth and Month
Depth
40
30
20
15
10
5
2
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Month
Temp (C)
15
10
5